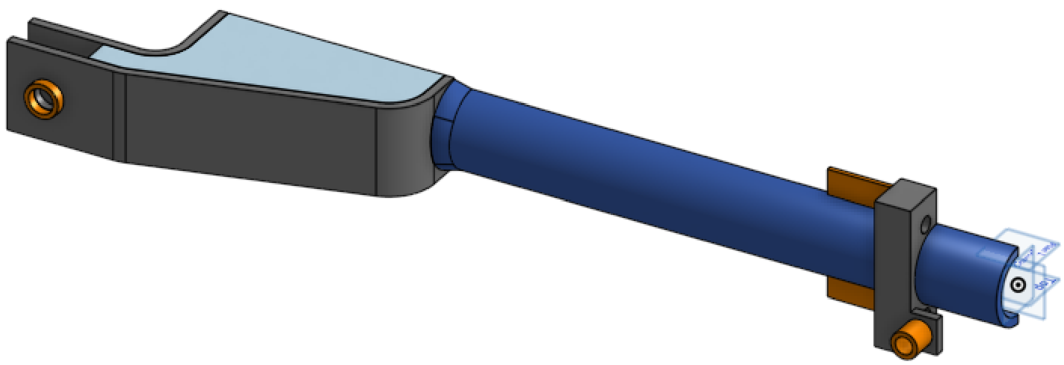
STATIC AND FATIGUE SIZING OF THE CRADLE

This is a censored document, no visual of the craddle (CAD, FEA) nor numerical values (graphes, results) are available here. The only goal of this document is to display my ability to conduct Finite Elements Analysis on a mechanical part.



For reference, here is a simplified model of the part on which FEA will be conducted. It doesn't correspond exactly to the tested part but it will give you a rough idea of what it should look like.

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General Presentation:

Define the context and scope of the calculation.

This study is the direct continuation of the study of the welded cradle carried out with shell elements. Using a volumetric model will provide better modeling of the weld zones. The goal here is to size the cradle on a closed geometry. This will determine the plate thicknesses of the welded assembly and identify the weld/bending areas. Sizing will be performed through successive iterations of the following process: CAD model, material, FE analysis, until a solution meeting the criteria is found. An analytical fatigue study over N cycles will then be performed to verify weld strength. We initially consider the following weld zones (see also: Figure: CAD model of the half-volumetric cradle):

Figures: envisaged weld zones (Orange lines on the CAD model)

Applicable standards and regulations.

Eurocode 3 / IIW, weld quality class FAT 50 (stress concentration factor 4 on T and fillet welds), because the welding processes considered are: root pass by TIG (GTAW) for controlled fusion, filling and finishing passes by MIG/MAG with low-hydrogen consumables.

Choice of sizing criterion.

We choose the material yield strength as the sizing criterion. It is possible that the part may plastify in certain areas without this being restrictive. The impact of this plasticization will be analyzed later via a fatigue study.

Ultimately, no plasticization.

Hypotheses and Input Data:

CAD model and input data.

We use a CAD model of the volumetric cradle (≠ shells). Recoiling mass: M tons. Load generated during firing: F kN, uncertainty x% → load used for FE analysis: 1.5*F kN.

From video analysis, we obtained a maximum force of F kN. (It will be confirmed in post-calculation (02/04/2025) that the maximum force at the interface with the dampers is F kN during firing.)

The input values used to calculate the recoil velocity were found online; we therefore estimate for each a 5% uncertainty u. With uncertainty propagation for multiplication, we obtain a recoil velocity uncertainty of $u_{vreculante,0}$ = x%.

However, the largest uncertainties come from the fact that operation occurs in a critical regime, and especially that all energy is converted into potential energy, which will appear in the Python PFD solution, in addition to uncertainties carried by the calculation of K and a (relative uncertainty as above: uK = x1% and uA = x2% excluding assumption uncertainty). Assuming 5% uncertainty for these assumptions, we obtain uK = x3% and uA = x4%. Using absolute uncertainty in an addition as in applying the PFD, the uncertainty u_{effort} = F kN is x6%. Our error factor is therefore x.

We will therefore retain a maximum force of 1.5*F kN.

Figure : CAD model of the half-volumetric cradle

Definition of calculation conditions. Modeling assumptions. Simplified geometry / structural assumptions. Modeling choice.

The calculation will be performed on the CAD model. We will conduct an elastic finite element analysis using ABAQUS, followed by an analytical fatigue study on critical zones. The number and location of critical zones may evolve throughout the iterative process.

Boundary conditions. Symmetry assumptions if applicable.

Pivot joint (sliding pivot + 1 node locked in translation) with X-axis at the interfaces with the mount. At the interface with the equilibrators: translation blocked along X and Z and rotation blocked along Y and Z. Transverse plane symmetry (plane normal to X).

Applied loads, static loads.

No data regarding the holding dampers (between mount and cradle): joint modeled as a slider with the frame. Static pressure due to the weight of the recoiling mass on the inside of the cradle. No CAD model of the recoiling mass provided.

Figure: load case (red: cradle; blue: recoiling mass; green: elevation/equilibrator actuators)

Material used. Mechanical properties: Young's modulus (E) Poisson's ratio (ν) Yield strength (Re) Tensile strength (Rm)

Steel 15CrMoV6

E = 210,000 MPa

ν = 0.3

Re = σ MPa

Rm = σ' MPa

It is machinable, weldable, and formable (https://www.aubertduval.com/wp-content/uploads/2024/12/SCV_FR.pdf). Available in sheets of thickness e mm and e' mm from a French supplier funded by the government.

Figure: application of 15CrMoV6 steel

Calculation Methodology:

Presentation of the calculation approach.

Static finite element analysis followed by analytical fatigue analysis depending on the number and layout of plastification zones.

Mesh convergence verification.

To have at least two elements across plate thickness, the maximum element size is e mm (geometric criterion). We will therefore aim to use e mm tetrahedral elements. The geometry is complex, requiring tetrahedral elements (structured/unstructured hex mesh impossible). These are well adapted to problems with high stress gradients or curved shapes. Because the CAD cradle model is "heavy," we will verify mesh convergence on a simplified geometry with a known analytical solution: a cylinder in compression (L= ; dint= mm; dext= mm; F= kN). Results for meshes e1 mm, e2 mm, e3 mm:

Figure: cylinder in compression in Abaqus

Analytical solution: σ = σ MPa

Standard deviation: x%, acceptable. These elements can be used.

Finite element modeling. Type of elements.

Use of 10-node tetrahedral elements with characteristic size e mm.

Figure: mesh of the craddle in Abaqus

Mesh convergence: e1, e2, e3 mm in Abaqus

Figure: mesh convergence (e1, e2, e3 mm) in Abaqus

Software assumptions. Representativeness of modeling vs real world.

Small-strain assumption (<5%). Material homogeneity (same mechanical properties in plates and weld beads).

Calculation Results:

Presentation and interpretation of results.

Maximum Von Mises stress: σ1 MPa; however, only one element exceeds σ2 MPa, located at the equilibrator arm attachment. Most elements lie between σ3 and σ4 MPa. Maximum displacement: x1 mm. Reduction of clearance with recoiling mass: x2 mm.

Figures: craddle stress (Von Mises) in Abaqus

Figure: craddle displacement along Z-axis in Abaqus

The diameter of the recoiling mass increases by x3 mm due to gas pressure, and a sliding fit (H7g6) must be maintained during firing.

Figure: sliding fit (www.H7g6.fr)

Thus, the minimum clearance needed is ...+…+... = j mm. We do not know the exact diameter of the recoiling mass, but the bore diameter must be at least d + j mm. For the mean and maximum clearances, this gives d + j_m mm and d + j_max mm respectively.

Stress analysis in the envisaged weld zones

A separate verification is needed in weld zones due to stress concentration factors.

Figure: weld layout (Orange lines on a CAD model of the craddle)

Reminder: T-welds in zones 1 and 2, fillet welds in zones 3 and 4. FAT 50 → stress concentration factor 4.

Figures showing weld stresses...

Maximum stress is σ5 MPa in a specific zone. Most stresses are around σ6 MPa. Same for welds fixing pivots at interfaces (not shown).

Nuances to the results

Additional stiffness exists due to unintended rigid connections between the tubular part and top/bottom plates. A sectional view (normal Y) shows fusion of adjacent solid parts during thickening—unlike the shell-element study where plates were not directly connected to the tube.

Figure: sectional view of the craddle showing a misrepresentation of some of the welded joints

Fusion occurs in the red zones. We kept this model for its increased rigidity and added two additional weld zones.

Stress in new weld zones (red)

Figures: weld bead 1, bead 2

Von Mises stress does not exceed σ7 MPa.

Verification and Validation:

Comparison with sizing criteria.

Criterion: Re = σ' MPa

Flat/bent zones: σ5 MPa < σ' MPa

Welds (stress factor 4):

Weld 1: 4*σ2 MPa < σ' MPa

Weld 2: 4*σ3 MPa > σ' MPa

Weld 3: 4*σ4 MPa < σ' MPa

Weld 4: σ6 MPa * 4 far exceeds yield and tensile strength.

Comparison with allowable limits including safety factor.

Flat/bent zones: safety factor 4

Weld 1: safety factor 1.3

Weld 2: N/A

Weld 3: safety factor 1.5

Weld 4: N/A

Iteration of the Process:

Modifications:

Addition of stiffener near zone 2. Increase of bending radius near weld 4. Plate thickness cannot be increased due to bending radius constraints.

Change of weld type to full penetration (FAT 80 or 90 → stress factor 2.5).

Results:

Figures: stresses and displacements

Stress in weld 4

Stress in bevel edges (weld 2, mid)

Verification and Validation:

With stress factor 2.5:

Weld 4: 2.5*σ7 MPa < σ’ MPa

Weld 2: 2.5*σ8 MPa < σ’ MPa

No plasticization.

Safety factors:

Weld 4: 1.5

Weld 2: 3.3

Conclusion and Recommendations (elastic part):

Summary:

This study determined static stresses and displacements in the volumetric cradle subjected to a load of F kN (>… kN). A plate thickness of e mm satisfies elastic criterion σ' MPa (15CrMoV6) for most elements. Geometric modifications were required to validate criteria in the inner weld and middle circular weld. Only one element plastifies, not in a critical area. No risk of jamming; radial deformation in tube on order of hundredths of a mm.

Constraint compliance:

Elastic limit criterion satisfied. Safety factors mostly >1.3.

Limitations:

Fatigue strength must be verified. Top and bottom plates must be machined before welding to the tubular section.